# Mini Banking System using Layered Architecture

*Problem statement:*

- Simulation of core banking operations
- Designed using OOP principles

Target User:

- Students / simulation

# TITLE: SYSTEM ARCHITECTURE

**Layered Architecture:**

Presentation Layer → User interaction (Main.java)

Service Layer → Business logic (AccountService)

Repository Layer → Data handling (AccountRepository)

**Key Classes:**

Account (base class)

SavingsAccount, CheckingAccount

**Package Structure:**

model | service | repository | ui

# OBJECT-ORIENTED DESIGN

- **Encapsulation**
  → Protects account balance using private variables
- **Inheritance**
  → Savings & Checking extend Account
- **Polymorphism**
  → withdraw() behaves differently
- **Abstraction**
  → Service layer hides internal logic

# Key Design Decisions

- **Efficient Data Storage**
  → HashMap used for fast access (O(1))
- **Separation of Concerns**
  → Clear division between UI, logic, and data
- **Input Validation**
  → Prevents invalid transactions
- **Flexible Design**
  → Easily extendable (GUI / Database)

# System Demonstration

- Create new account
- Deposit money
- Withdraw amount
- Error handling (insufficient balance)
- Transfer funds